



OPEN Adaptive reinforcement learning framework for sustainable microgrid optimization in arid urban environments

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Managing energy in urban microgrids is a major issue because of the high degree of variability of renewable energy sources and the dynamic nature of the urban demand especially in regions with arid climates that pose extreme temperatures that provide volatile cooling loads. Existing energy management systems (EMS) often relied on static or rule-based control systems, which lack the responsiveness needed to manage the variability inherent in renewable generation and fluctuating demands. This study presents a simulation-based and adaptive reinforcement learning (RL)-based energy management framework that addresses persistent inefficiencies in coordinating diverse energy sources within urban microgrids, particularly in arid regions, to bridge this gap. In this study, it was tried to develop a simulation-driven platform combining EnergyPlus with Python/TensorFlow RL agents to dynamically optimize the dispatch of solar, wind, diesel, and battery resources. Unlike prior approaches, the proposed system also combined inter-microgrid communication via MQTT protocols, enabling real-time energy sharing. The framework was validated in a case study reflecting Riyadh's climatic conditions, where it significantly improved operational and environmental performance. Simulation outcomes demonstrated high predictive accuracy for hourly and annual consumption ($R^2 = 0.94$ and 0.90 , respectively). Compared to baseline methods, the RL-based approach reduced CO_2 emissions by 14%, SO_2 emissions by 13.6%, and primary energy consumption by 10%.

Keywords Cost efficiency, Energy management, Microgrid optimization, Reinforcement learning, Renewable energy

Abbreviations

AI	Artificial intelligence
BESS	Battery energy storage system
DQN	Deep Q-network
EMS	Energy management systems
GHG	Greenhouse gas
IPCC	Intergovernmental panel on climate change
IoT	Internet of Things
LCA	life cycle assessment
LCI	Life cycle inventory
MAE	Mean absolute error
RMSE	Mean squared error
MQTT	Message queuing telemetry transport
NZEB	Net zero energy buildings
NMG	Networked microgrid

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NMBE	Normalized mean bias error
P2P	Peer-to-peer
PV	Photovoltaic
RL	Reinforcement learning
SOC	State of charge
ToU	Time-of-use

List of symbols

S_i	Simulated value at data point i
M_i	Measured value at data point i
N_p	Number of data points
M	Mean of the measured values
I	Solar irradiance
A_{solar}	Area of solar panels
ρ	Air density
A	Rotor swept area
v	Wind speed
C_p	Power coefficient
P_{solar}	Solar power output
P_{wind}	Wind power output
F_{diesel}	Diesel generator fuel consumption
P_{diesel}	Power output of the diesel generator
η_{diesel}	Efficiency of the diesel generator
F_{diesel}	Diesel fuel consumption
Fuel Cost	Cost of the diesel fuel
C_{diesel}	Operational cost of diesel generator
P_{import}	Power imported from the grid
P_{export}	Power exported to the grid
P_{demand}	Power demand
P_{wind}	Power generated by wind turbines
P_{solar}	Power generated by solar panels
P_{diesel}	Power provided by the diesel generator
P_{storage}	Power provided by energy storage
$C_{\text{grid}}(t)$	Instantaneous operational cost at time t
$LPSP_{\text{penalty}}(t)$	Loss of power supply probability penalty
$P_{\text{curtail}}(t)$	Penalty for curtailed renewable energy
ω_1	Weighting factor for economic goals
ω_2	Weighting factor for reliability
ω_3	Weighting factor for sustainability
$Q(s_t, a_t)$	Q-value at state s_t and action a_t
α	Learning rate
γ	Discount factor
R_t	Reward function for RL
$P_{\text{load}}(t)$	Power load demand at time t
$P_{\text{wind}}(t)$	Power from wind at time t
$P_{\text{battery}}(t)$	Power from battery at time t
$P_{\text{grid}}(t)$	Power from the grid at time t
Δt	Time step (usually hourly)
EF	Emission factor
K_{PV}	Utility gain of the PV inverter
T_{PV}	Time constant of the power electronic interface
K_w	Wind power conversion efficiency gain
T_w	Time constant of the wind turbine system
K_D	Gain of the diesel generator
T_g	Governor time constant
T_d	Engine time constant
K_B	Gain of the bi-directional power converter
T_b	Time constant of the battery interface

Recent improvements have established NZEBs as essential elements in renewable energy communities, facilitating a transition from self-sufficiency to energy sharing models to address the growing need for sustainable energy systems¹. NZEBs attain energy neutrality via on-site renewable sources and bolster grid stability by facilitating the distribution of surplus energy within microgrids, hence enhancing overall efficiency and resilience. Energy-sharing frameworks within microgrids, bolstered by NZEBs, enable balanced energy distribution, diminish reliance on the central grid, and mitigate the intermittency issues linked to renewable energy^{2–4}. This progression links NZEBs with the overarching objectives of renewable energy communities, wherein decentralized power generation and adaptable load management foster sustainable, community-oriented energy solutions that enhance microgrid stability and optimize energy usage^{5,6}. However, the successful combination of these buildings into a larger system necessitates the development of microgrids.

The growing demand for sustainable and reliable energy solutions has led to the development of microgrids, which are small-scale power grids operate independently or in conjunction with the main electrical grid^{7,8}. Reflecting this trend, studies have explored different configurations for near-autonomous operation. Optimal sizing and configuration of hybrid renewable energy systems, including PV, wind are needed to enhance cost-efficiency and atmospheric emissions minimization in urban regions⁹. Stamatellos and Stamatellou¹⁰ explored a novel approach to establishing near-autonomous microgrids in commercial and public buildings, focusing on achieving a nearly ZEB model. They discussed integrating both short and long-term energy storage solutions to optimize micro-grid operation. Short-term storage was facilitated by large batteries or employees' electric vehicle batteries in bi-directional charging mode, while long-term storage relied on hydrogen, produced through electrolysis using surplus electricity from PV systems. The stored hydrogen was utilized as fuel for a hydrogen spark ignition engine connected to a synchronous generator, allowing flexible energy production. A size optimization study was conducted to determine optimal ratings for the genset engine, electrolyzer capacity, and the rooftop PV panels' tilt angle. The findings recommended a 40 kW genset engine, a 39° tilt angle for PV panels, and a maximum electrolyzer capacity of 72 kW.

In addition to storage-based designs, microgrids are becoming more and more dependent on the synchronized work of various sources of energy. Microgrids are essential for combining conventional power generation techniques, such diesel generators and energy storage systems, with renewable energy sources, like solar and wind^{11,12}. Efficient energy utilization in microgrids is vital for maintaining a stable and continuous power supply, especially due to the fluctuations of renewable energy sources^{13,14}. Alam et al.¹⁵ provided a systematic review of the current challenges and opportunities associated with combining renewable energy sources, particularly solar PV, into DC microgrids. DC microgrids have gained considerable attention for their potential to improve energy efficiency and support renewable energy integration. However, large-scale combining of renewables in these systems presents significant challenges, including issues with voltage regulation, energy management, inertia control, and managing uncertainties. Various control methods (droop control, centralized control, distributed control, virtual inertia control, and uncertainty management algorithms) have been proposed to tackle these challenges, yet a comprehensive overview of these approaches and future trends has been lacking.

Given these technical complexities, effective energy management becomes central to microgrid performance. Enhancing energy management in microgrids improves operating efficiency, maximizes renewable energy utilization, decreases dependence on fossil fuels, and saves energy expenses^{16–18}. Efficient energy management solutions are crucial for preserving the stability and reliability of microgrids, thereby guaranteeing that energy demands are regularly and sustainably fulfilled^{19,20}. In this regard, Kumar et al.,²¹ explored advanced techniques in machine learning to develop a robust monitoring and management system, aiming to optimize energy consumption and improve the overall efficiency and profitability of microgrids. In another study, Rodriguez-Gil et al.²² examined the role of EMS in optimizing the efficiency and reliability of networked microgrids, which have become increasingly important for integrating renewable energy resources and enhancing grid resilience. They provided a comprehensive overview of EMS in NMGs, covering essential aspects such as system architecture, optimization algorithms, control strategies, and the combining of distributed energy resources. Their paper addressed the challenges and opportunities within energy management for NMGs and reviews key approaches and technologies being utilized. Additionally, it discussed the impact of emerging trends on the design and performance of EMS in NMGs. Through this extensive analysis, their paper aimed to equip researchers, practitioners, and policymakers with insights into current developments and future directions in EMS for networked microgrids.

Nevertheless, traditional EMS methods even with such progress do not cope with dynamic and uncertain operating conditions. Despite the advances in microgrid technology, the dynamic nature of energy production and consumption presents significant challenges in achieving optimal energy management²³. Traditional monitoring systems often fall short in addressing the complexities associated with integrating diverse energy sources, predicting consumption patterns, and ensuring efficient distribution of power²⁴. There is a pressing need for a novel monitoring system that can handle these intricacies with high accuracy and adaptability.

This requirement has promoted more advanced paradigms to control. Current methods and systems for monitoring and managing microgrid energy consumption encompass a diverse range of approaches, each with its own strengths and limitations. Traditional systems often rely on rule-based algorithms and static optimization techniques, which, while effective in stable conditions, struggle to adapt to the dynamic nature of microgrids with high penetration of renewable energy sources²⁵. These conventional methods typically use predefined schedules and heuristic strategies to manage the dispatch of energy from various sources, aiming to minimize costs and ensure reliability. Supervisory control and data acquisition systems are widely used to provide real-time monitoring and control, enabling operators to make informed decisions based on current grid conditions²⁶. However, these systems can be limited by their dependence on historical data and their inability to predict and adapt to future uncertainties in load and generation. Additionally, advanced control strategies, such as model predictive control, have been employed to enhance the operational efficiency of microgrids by optimizing control actions over a finite horizon, taking into account future predictions of load and generation²⁷. Despite these advancements, the combination of heterogeneous energy sources and the need for real-time, adaptive management remain significant challenges. To overcome these rigidities of conventional control, researchers are increasingly turning towards more sophisticated, data-driven methods. Consequently, there is an increasing interest in more sophisticated methods, such as machine learning-based approaches, which offer the potential to address these challenges through improved prediction accuracy, adaptability, and decision-making capabilities²⁸.

Machine learning has increasingly been applied to the field of energy management, offering transformative potential through its ability to handle complex, dynamic environments²⁹. RL is one of the machine learning methods that have become especially promising in the case of complex energy systems. Recent developments have demonstrated that RL algorithms have a high level of success in managing the uncertainty of renewable energy

sources by adjusting parameters of the controller to keep the system stable³⁰. RL has emerged as particularly promising due to its capacity for learning optimal policies through interaction with the environment³¹. Previous applications of RL in energy management have demonstrated significant improvements in efficiency and adaptability. For example, RL has been employed to optimize the scheduling and dispatch of distributed energy resources, where it learns to make decisions that balance supply and demand while minimizing operational costs³². This approach has been particularly effective in handling the variability and uncertainty inherent in renewable energy sources such as solar and wind³³. Additionally, RL algorithms have been used to enhance demand response strategies, where the system learns to adjust consumer load in response to price signals or grid conditions, thereby reducing peak demand and improving grid stability^{34,35}. Studies have also explored the use of RL for energy storage management, enabling more effective utilization of batteries and other storage devices by learning optimal charging and discharging patterns. These applications highlight the versatility and potential of RL in creating adaptive, efficient, and resilient EMS. By utilizing the capacity of RL to perpetually enhance its performance through acquisition of knowledge, these systems can attain superior operational efficiency and dependability relative to conventional methods³⁶. Machine learning, particularly RL, offers a promising solution in this regard. By utilizing extensive data, machine learning algorithms may discern and forecast consumption trends, optimize resource distribution, and improve the decision-making process^{37,38}. The implementation of a machine learning-based monitoring system can thus revolutionize the management of microgrids, making them more efficient, reliable, and sustainable³⁹. However, there are still significant gaps in the literature of RL-based microgrid management.

Despite the advancements in applying machine learning to energy management, several gaps remain in the existing research that the proposed system seeks to address. One significant gap is the limited ability of current systems to adapt to the highly variable and uncertain nature of energy supply and demand in microgrids⁴⁰. Traditional methods and even some machine learning applications often struggle to integrate diverse energy sources dynamically and efficiently⁴¹. Arévalo et al.⁴² provided a systematic review of the role of microgrids in advancing sustainable and resilient energy systems, focusing on the integration of renewable energy sources and decentralized energy management. Using the PRISMA methodology, the authors analyzed 74 peer-reviewed articles out of 4205 studies published between 2014 and 2024. The review explored essential areas for enhancing microgrid efficiency and reliability, including RL, multi-agent systems, predictive modeling, energy storage, and optimization algorithms. It also highlights emerging technologies such as AI, the IoT, and flexible power electronics, which offer significant potential to improve energy management and operational performance in microgrids. Despite these advancements, challenges remain, particularly in incorporating AI within complex, real-time control systems and effectively managing distributed energy resources. Their paper identified critical research opportunities aimed at enhancing microgrid scalability, resilience, and efficiency, emphasizing their crucial role in the transition to sustainable energy solutions.

Further analysis of the recent literature also demonstrates the similar methodological and operational constraints. Recent studies reveal several recurring limitations in microgrid energy management research. While deep RL has significantly improved adaptive real-time control capabilities, most existing frameworks remain confined to single-microgrid or component-level optimization, with limited emphasis on coordinated operation across multiple microgrids. Furthermore, uncertainty handling is often implicit or partially addressed, and environmental performance is typically evaluated as a post-processing step rather than being integrated into the control framework. Table 1 shows the comparative synthesis of microgrid energy management studies.

In continuation of the earlier developments in the area of microgrid energy management, there has been great advancements in the areas of rule-based control strategies and optimization-based strategy which has been able to solve the issues of energy dispatch and operational control in single systems among the microgrids. These approaches, however, are mostly deterministic-based and they do not have the effectiveness to assess uncertainty and environmental factor.

The suggested system utilizes RL to establish a more flexible and responsive management framework, proficient in optimizing energy allocation in real-time. Additionally, many existing approaches lack a comprehensive strategy for incorporating both renewable energy sources and conventional generators, often focusing on one at the expense of the other⁵¹. The suggested system resolves this issue through combining diesel generators with renewable sources such as wind and solar, so assuring a balanced and reliable energy supply.

Study	Control paradigm	Adaptive real-time control	Multi-microgrid coordination	Uncertainty handling	Environmental assessment	EMS based evaluation
⁴³	Machine learning-based	Yes	No	Yes	No	No
⁴⁴	Machine learning-based	No	No	Yes	No	No
⁴⁵	Regression learner-based	No	No	Yes	No	No
⁴⁶	AI and Nanomaterials-based	No	No	Yes	Yes	Yes
⁴⁷	DRL-based	Yes	Yes	Yes	Yes	Yes
⁴⁸	Deep Learning-based	Yes	No	No	No	Yes
⁴⁹	AI-based	Yes	No	Yes	No	Yes
⁵⁰	Deep Learning-based	Yes	No	Yes	No	No
Current Study	RL-based	Yes	Yes	Yes	Yes	Yes

Table 1. Comparison of recent approaches to microgrid energy management.

Furthermore, while previous works have made strides in predictive accuracy, they frequently fall short in real-world implementation fidelity and the incorporation of reliability factors. The proposed system aims to enhance prediction accuracy and operational fidelity, ensuring that load uncertainties are effectively managed and that the system remains robust under various conditions. Moreover, precise cost assessment and economic viability are areas that have not been fully optimized in earlier studies. The proposed system innovates by including detailed expense assessments, which improve the overall economic efficiency of microgrids. This comprehensive strategy improves operational efficiency and boosts the profitability of EMS.

Based on shortcomings in Table 1, namely that the current RL models do not involve coordination and integration of multi-microgrids with the environment, this research attempts to address these gaps. Despite RL having demonstrated great potential in managing microgrid energy, the majority of the literature has been constrained to isolated or component-scale control and does not involve the coordinated behavior in the case of realistic and uncertain operating environments. Specifically, existing RL models are scarcely designed to consider joint benefits of operational cost, system reliability, and environmental performance in a single control system. Additionally, most of the past research presumes idealistic system behaviors without considering the practical limitations like load uncertainty, the reliability of generators, dispatchability and fidelity of real-time implementation. Economic performance is not considered as a primary optimization objective and often treated as a secondary outcome. The motivation for this study stems from the limitations of traditional microgrid management, which primarily relies on rigid, rule-based controllers that struggle with the stochastic nature of modern energy environments. To address these challenges, a simulated monitoring and management system that uses RL is offered to optimize energy consumption and improve system performance. This system integrates diverse energy sources, including renewable options (wind and solar), conventional diesel generators, and energy storage solutions, to effectively model complex energy dynamics. The RL model is designed to handle load uncertainties with high precision while embedding advanced reliability factors and detailed cost assessments to improve both operational efficiency and economic viability. The specific objectives of this study are to:

- (i) Develop an RL-based control framework that dynamically optimizes energy distribution in microgrids.
- (ii) Integrate heterogeneous energy sources within a simulated environment to evaluate system performance under variable conditions.
- (iii) Compare the proposed system with traditional baseline methods (fixed energy dispatch, static load balancing, and conventional cost minimization) to demonstrate improvements in energy efficiency, cost reduction, and profitability.
- (iv) Establish a scalable and robust framework that can pave the way for future microgrid technologies and real-world applications.

The remainder of this paper is organized as follows: Sect. 2 describes the system design, simulation framework, and detailed methodology, including the RL model implementation and the integration of diverse energy sources. Section 3 presents the simulation results, including a comparative analysis between the proposed RL-based approach and traditional baseline methods, along with discussions on performance metrics. In addition, Sect. 3 provides a comprehensive profitability and efficiency analysis, highlighting the economic and operational benefits of the proposed system. Section 4 concludes the study by summarizing the key contributions, discussing limitations, and suggesting directions for future research.

Methodology

Hypothesis, assumptions, and research questions

This study was based on the hypothesis that combining RL into microgrid management systems could significantly enhance energy distribution efficiency, reduce operational costs, and improve overall system performance compared to traditional control methods. It was hypothesized that an RL-based management system, due to its ability to dynamically adapt to load fluctuations and renewable variability, would outperform conventional rule-based and static optimization approaches. To build a reliable simulation model, the following assumptions were made:

- The EnergyPlus simulation accurately represented real-world microgrid operations in Riyadh, capturing seasonal and daily variations in energy demand and renewable energy generation.
- The performance characteristics of all simulated components—such as solar panels, wind turbines, diesel generators, and battery storage—were based on standardized data and operate under near-ideal conditions, with minor deviations managed via probabilistic models.
- The cost parameters and operational metrics (e.g., energy costs, fuel consumption, and maintenance expenses) used in the simulation were reflective of current industry standards.
- The simulated communication protocols (e.g., MQTT) and control mechanisms accurately emulated real-time data exchange and decision-making processes within microgrid systems.
- Traditional EMSs, characterized by fixed energy dispatch, static load balancing, and conventional cost minimization, served as a valid baseline for comparative analysis.

The study is guided by the following primary research questions:

- Can an RL-based management system optimize energy distribution more effectively than traditional rule-based control methods?

- Does the integration of heterogeneous energy sources (renewable, diesel generators, and energy storage) within the simulated environment lead to measurable improvements in energy efficiency and operational cost reduction?
- What are the environmental and economic benefits—specifically in terms of reduced carbon emissions and increased net profitability—of adopting the RL-based approach compared to conventional methods?
- How robust is the RL-based system in adapting to real-time fluctuations in energy demand and renewable energy supply over an extended simulation period?

System design and architecture

Building upon the hypothesis detailed in Sect. 2.1, this study employed a simulation-based analysis to evaluate the performance of an RL-based microgrid management system. The simulation framework was designed to optimize resource allocation and load management in real time—particularly under the variability of renewable energy sources such as solar and wind. The system dynamically adapted to fluctuations in energy supply and demand, aiming to reduce operational costs and enhance energy efficiency. Moreover, the simulation combined multiple energy sources, including renewables and diesel generators, to comprehensively assess the economic and environmental benefits of energy sharing across interconnected microgrids⁵².

The simulation system designed for monitoring and managing microgrid energy consumption was composed of both hardware and software components, meticulously integrated to achieve optimal performance. The system's architecture was developed with a focus on scalability, reliability, and data processing, ensuring seamless coordination between the microgrids and the neighborhood system (Fig. 1).

The simulation process was initiated by gathering a comprehensive primary dataset that included hourly building energy consumption profiles, historical temperature and climate records, global radiation levels, wind speeds, and fuel consumption rates for diesel generator. These input variables were derived from standardized benchmark models representative of urban buildings in Riyadh. EnergyPlus was employed to process these inputs and to simulate detailed energy flows, renewable generation curves, load profiles, and storage levels for the microgrid system. The outputs generated by EnergyPlus were then used to construct the state vector for the RL algorithm. This state vector comprised as: (i) Renewable generation values, (ii) Building load consumption, (iii) Battery state-of-charge (ranging from 20% to 100%), and (iv) Diesel generator operational status. Once the state vector was formulated, the RL algorithm (implemented using Python and TensorFlow) was trained to select optimal control actions. At each time step, the RL model received the state vector and determined a control action, which could include, (i) Dispatching energy from renewable sources, (ii) Commanding battery charging or discharging, and (iii) Activating or deactivating diesel generators. Following the action selection, the simulation environment was updated accordingly. The effect of each control action on the system's performance was measured using a reward function that considered factors such as the cost of energy production, load balancing efficiency, and renewable utilization. The reward feedback was used to update the RL policy via the Bellman equation in a manner consistent with the principles of Q-learning. This iterative loop—where the simulation provided state inputs, the RL algorithm selected actions, and a reward function evaluated the outcomes—was repeated over many training episodes. The simulation was conducted on two-time scales: (i) an hourly resolution was used to capture short-term dynamics such as daily load peaks and renewable intermittency, and (ii) an annual simulation period was employed to aggregate these effects and capture long-term trends, including seasonal variations in energy generation and demand. The final output of the simulation was an optimal control policy for managing energy distribution across the microgrid system. This policy was demonstrated by significant improvements in key performance metrics, such as increased energy efficiency, reduced operational costs, and enhanced profitability. The innovations in this study lay in the dynamic, RL-based approach that integrated multiple energy sources, facilitated inter-microgrid interactions via a realistic communication network (using MQTT), and effectively adapted to diverse energy scenarios—advantages that were not fully realized in traditional rule-based or static optimization methods. At the final step, to enhance the performance assessment of the RL-based microgrid management system, the LCA was simulated to measure the environmental impacts linked to microgrid operations.

Region of study and proposed microgrids system

Overview of Riyadh

This study focuses on the energy management needs of Riyadh, Saudi Arabia. Riyadh, with a population of approximately 7 million and an area of 1973 km², presents a unique and challenging environment for energy management due to its arid climate. The region experiences extreme temperature variations, which significantly influences energy demand patterns, particularly increasing the need for cooling during the hot summer months. Moreover, the city receives a high level of solar radiation, which offers substantial potential for solar energy generation. However, the average wind speed in the region is relatively low, limiting the feasibility of wind energy as a primary renewable resource. In the context of Riyadh's energy infrastructure, diesel generators are commonly used as backup sources, while battery storage systems are also employed. These parameters underscore the challenges faced in balancing energy supply and demand in a region where renewable energy sources, despite their abundance in solar potential, are intermittent and where conventional diesel generators impose higher operational costs and environmental impacts. The significance of this study lies in its focus on developing an RL-based microgrid management system that is tailored to address these specific challenges. The study's simulation framework, which combines the diverse energy scenarios and load profiles characteristic of Riyadh, is expected to provide valuable insights into how advanced control strategies can improve the economic and environmental viability of urban energy systems in arid regions.

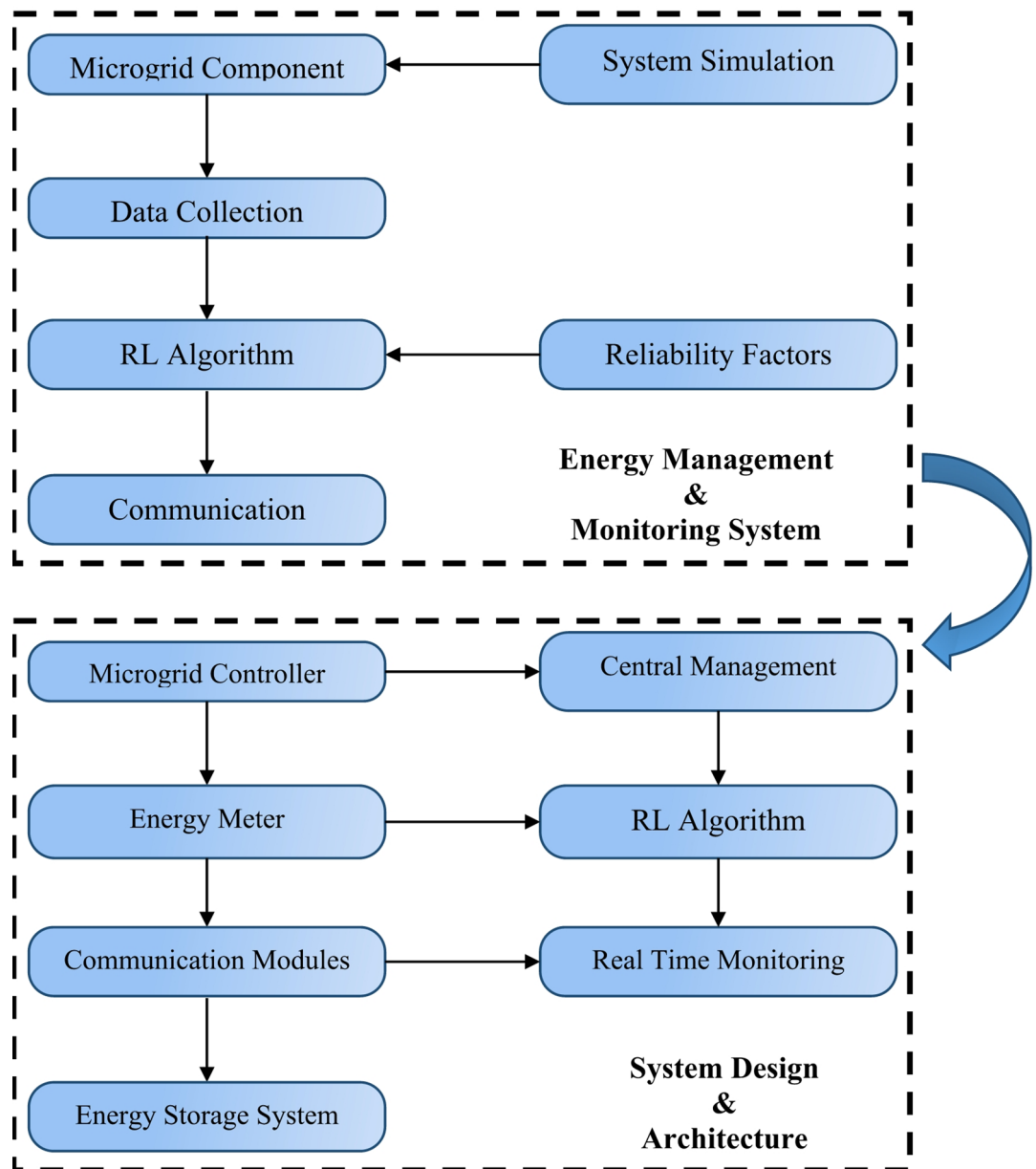


Fig. 1. Flowchart of the current proposed research method.

Microgrid system Configuration, Interactions, and diverse energy scenarios

As it was discussed, the microgrid is a local system of distributed energy sources (such as solar panels, wind turbines, diesel generators, and energy storage systems) and consumers (loads) that can operate independently or simultaneously with the main electricity grid. In other words, a microgrid is a small, self-sufficient network of energy sources and consumers whose management intelligently coordinates production and consumption. In the current study, the focus was on optimizing the performance of microgrids using the RL algorithm. For this reason, instead of examining the details of a specific building, it was tried to work on the simulation and management of microgrids. In fact, buildings were considered as input sources (load profiles); that is, energy consumption patterns were taken from standard buildings or benchmark models to provide realistic input data for the simulation. However, the main goal of the study is to investigate how to manage energy intelligently and optimally at the microgrid level, not to analyze the performance of a specific building. Five representative microgrids were modeled concurrently to reflect typical urban energy systems in Riyadh. The microgrids were configured with varying solar panel, diesel generator, wind turbine and battery⁵³. The five microgrids utilized in this framework were configured based on a spectrum of theoretical urban archetypes. As shown in Table 2, the sizing criteria for each microgrid reflect actual energy consumption benchmarks in Saudi Arabian urban districts. MG1 and MG2 represent residential profiles with higher battery-to-PV ratios to accommodate nighttime cooling demands. MG3 follows a mixed-use pattern, while MG4 and MG5 simulate commercial environments.

Microgrid ID	Profile type	Solar panel (kW)	Diesel generator (kW)	Battery (kWh)	Wind turbine (kW)
MG1	Residential (Large)	10	20	15	–
MG2	Residential (Small)	–	10	10	5
MG3	Commercial- Residential	5	–	–	5
MG4	Commercial	–	20	15	–
MG5	Commercial	5	15	15	5

Table 2. Characteristics of the simulated microgrids.

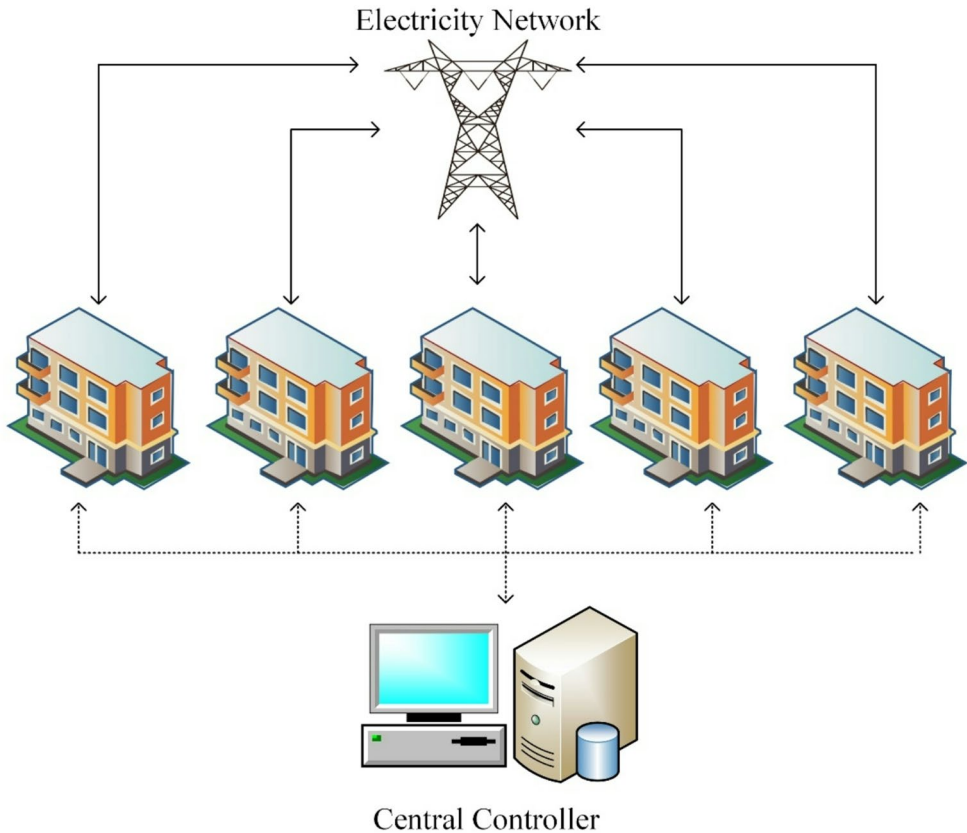


Fig. 2. Schematic diagram of the simulated microgrid system.

A schematic diagram was prepared to illustrate the overall configuration of the microgrid system (Fig. 2). Figure 2 illustrates the overall configuration of the simulated microgrid system. In the diagram, five representative microgrids (MG1-5) were modeled concurrently. Each microgrid is depicted as an independent node with its own renewable energy sources, diesel generators, and battery storage. Building load profiles (derived from benchmark models representative of urban buildings in Riyadh) and meteorological data serve as input for each microgrid. Inter-microgrid interactions are enabled via an emulated communication network (using protocols such as MQTT), which facilitates energy exchange among microgrids. Additionally, a bidirectional connection to an upstream network is shown, allowing for energy import and export as needed.

The decision-making process for inter-microgrid energy exchange is governed by the RL agent’s output in coordination with the MQTT communication layer. The RL framework explicitly optimizes the net energy position of the microgrid rather than a passive model. The agent follows a hierarchical priority. First of all, satisfying local load demand, then maintaining battery SOC above the critical threshold, and finally exporting surplus energy to the P2P network. In this study, a dynamic shadow pricing mechanism is used. When a microgrid has a deficit, it queries the MQTT broker for available surplus from neighboring nodes. If the internal cost of diesel generation or grid purchase exceeds the P2P clearing price (set at the mid-point between the feed-in tariff and the retail rate), a transaction is initiated.

Data collection

The primary dataset used in this study was assembled from multiple sources to provide realistic input values for the simulation of microgrid operations. The use of generalized datasets and simplified models is a common

Hour	Measured load (kW)	Simulated load (kW)	Hour	Measured load (kW)	Simulated load (kW)
0:00	61	72	12:00	50	61
1:00	45	51	13:00	54	60
2:00	55	50	14:00	60	55
3:00	55	70	15:00	45	39
4:00	56	51	16:00	30	39
5:00	46	48	17:00	42	40
6:00	50	46	18:00	58	60
7:00	35	41	19:00	59	65
8:00	60	70	20:00	65	60
9:00	45	42	21:00	60	72
10:00	45	43	22:00	50	45
11:00	60	50	23:00	60	78

Table 3. Comparison of measured vs. simulated hourly energy consumption for a representative day.

Metric	Value
RMSE	8.07 kW
MAE	6.92 kW
NMBE	5.00%

Table 4. Evaluation of measured and simulated hourly energy consumption.

practice in early-stage microgrid research, as it allows for assessing core system behaviors and the feasibility of proposed approaches before integrating complex, localized data. This approach is accepted in the literature, as noted by Schiffer et al.⁵⁴ and Nazir et al.⁵⁵, who emphasize the importance of using simplified models in initial evaluations to provide tractable analysis and understand system dynamics before proceeding to detailed modeling.

Hourly energy consumption data

Hourly load profiles were derived from benchmark models of urban buildings representative of Riyadh. These models were based on standard architectural parameters and occupancy patterns typically observed in the region. Consumption data were generated in kilowatt-hours and were observed to range between 30 and 70 kW for the various microgrids (Table 3).

To ensure the reliability and accuracy of these simulated load profiles, the data were validated through the comparison of the local energy consumption statistics and published benchmark data for similar urban environments. This validation process provided confidence that the hourly energy consumption data used as input for the RL algorithm were representative of actual urban energy usage in Riyadh. Statistical measures, including RMSE, MAE and NMBE were computed to assess the agreement between the simulated data and real-world observations. These three metrics collectively provided a comprehensive evaluation of the model's predictive performance. RMSE was chosen as it emphasizes larger deviations between measured and simulated values, thereby highlighting the impact of extreme errors Eq. (1)⁵⁶.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_i - M_i)^2} \quad (1)$$

MAE was selected to provide a straightforward average of the absolute differences, offering a clear indication of the overall error magnitude while being less sensitive to outliers Eq. (2)⁵⁷.

$$MAE = \frac{1}{N} \sum_{i=1}^N |S_i - M_i| \quad (2)$$

NMBE was incorporated to quantify the systematic bias in the simulation by normalizing the mean error relative to the average measured load, and it is expressed as a percentage for ease of interpretation Eq. (3):

$$NMBE = \frac{\frac{1}{N} \sum_{i=1}^N (S_i - M_i)}{\bar{M}} \times 100 \quad (3)$$

The evaluation of the measured and simulated hourly energy consumption data yielded an RMSE of 8.07 kW, an MAE of 6.92 kW, and an NMBE of 5.00%. These values indicated that the simulation produced reasonably accurate results. Given that the typical load levels in this microgrid simulations, an error margin of around 8

kW is relatively modest. Furthermore, the NMBE of 5.00% shows only a slight positive bias, suggesting that the simulation did not systematically overestimate or underestimate the consumption. Therefore, these metrics collectively confirm that the simulated energy consumption data are reliable for our study and provide a robust basis for evaluating the RL-based microgrid management system.

Temperature and climate records

Historical climate data were obtained from regional meteorological databases and included daily temperature readings, solar irradiance, and wind speed measurements (Fig. 3).

The data indicated that average summer temperatures were approximately 43.7°C and winter temperatures around 25.6°C. Global radiation values were recorded at about 7585.1 MJ.m⁻².year⁻¹, while wind speeds ranged from 1.1 to 4.5 m.s⁻¹. These records were cross-validated with global climate datasets (such as those provided by the World Meteorological Organization) to ensure consistency with local conditions.

Fuel consumption rates for diesel generators

Technical specifications for diesel generators were obtained from manufacturer data and regional energy reports. In this study, the diesel generator capacities for the simulated microgrids were selected to reflect typical configurations for urban microgrid systems in Riyadh. As shown in Table 2, MG1 and MG4 were assigned a diesel generator capacity of 20 kW, MG2 a capacity of 10 kW, and MG5 a capacity of 15 kW, while MG3 was configured without a diesel generator. Fuel consumption rates were calculated using established performance Eqs^{58,59}.

Energy sources integration and interactions

A comprehensive energy sources integration framework was developed to simulate the operation of urban microgrids in Riyadh. The simulation incorporated multiple energy sources include renewable systems (solar panels and wind turbines), diesel generators, and energy storage devices, along with an upstream network connection for energy exchange. The approach was designed to capture the variability of renewable generation, the reliability of conventional sources, and the economic impacts associated with each component⁶⁰. Renewable energy generation was modeled using standard equations. Solar panel output was calculated as follows Eq. (4)⁶¹.

$$P_{\text{solar}} = I \times A_{\text{solar}} \times \eta \quad (4)$$

Wind turbine output was determined by Eq. (5):

$$P_{\text{wind}} = \frac{1}{2} \rho A v^3 C_p \quad (5)$$

Historical and real-time climate data were obtained from regional meteorological databases and cross-validated with global datasets to ensure consistency with Riyadh's conditions⁶². Energy storage was incorporated to manage renewable intermittency. The state-of-charge of batteries was updated using Eq. (6)⁶³.

$$\text{SOC}_{t+1} = \text{SOC}_t + \frac{P_{\text{charge}} \times \eta_{\text{battery}} - P_{\text{discharge}}}{C_{\text{battery}}} \quad (6)$$

Diesel generators were integrated as a backup power source to ensure reliability during periods of low renewable generation or high demand. Unlike generic values sometimes reported, diesel generator capacities were tailored to the simulated microgrid configurations (see Table 2). Fuel consumption rates were computed as Eq. (7)⁶⁴.

$$F_{\text{diesel}} = \frac{P_{\text{diesel}}}{\eta_{\text{diesel}}} \quad (7)$$

The operational cost was estimated by Eq. (8).

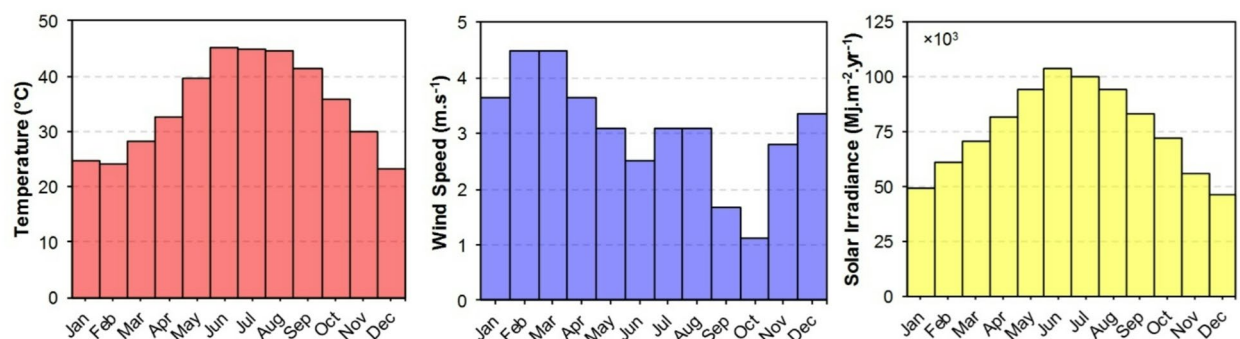


Fig. 3. Climatic data of the Riyadh over a year.

$$C_{\text{diesel}} = F_{\text{diesel}} \times \text{Fuel Cost} \quad (8)$$

These specifications were validated against manufacturer data and regional reports, and further confirmed using operational data from local energy providers.

A bidirectional connection to an upstream network was modeled to provide additional energy flexibility. This connection allowed for both import and export of energy, governed by the equations Eq. (9)⁶⁵.

$$P_{\text{import}} = P_{\text{demand}} - (P_{\text{wind}} + P_{\text{solar}} + P_{\text{diesel}} + P_{\text{storage}}) \quad (9a)$$

$$P_{\text{export}} = (P_{\text{wind}} + P_{\text{solar}} + P_{\text{diesel}} + P_{\text{storage}}) - P_{\text{demand}} \quad (9b)$$

Dynamic modeling of generating units and microgrids

To accurately simulate the response of the microgrid, each generating unit is modeled using its respective equivalent circuit. The grid-connected microgrid structure is represented using a block-level architecture consisting of generation units, power electronic converters, energy storage, local loads, and grid interconnection interfaces. A schematic architecture of the Microgrid is shown in Fig. 4. The PV system is modeled as a voltage-controlled source where the output power depends on irradiance and temperature. The dynamics of the DC/AC inverter are represented by a first-order transfer function Eq. (10).

$$G_{PV}(s) = \frac{K_{PV}}{1 + sT_{PV}} \quad (10)$$

The transfer function for the diesel generator unit is given by Eq. (11).

$$G_{DG}(s) = \frac{K_d}{(1 + sT_g)(1 + sT_d)} \quad (11)$$

The transfer function for the wind turbine unit and the dynamic response of the battery during charging/discharging is modeled as Eqs. (12) and (13) respectively.

$$G_W(s) = \frac{K_W}{1 + sT_W} \quad (12)$$

$$G_{BESS}(s) = \frac{K_B}{1 + sT_B} \quad (13)$$

In this study several nonlinearities were incorporated into the system model. These include the nonlinear efficiency of the diesel generators, which vary with the load ratio, and the nonlinear charging/discharging characteristics of the BESS as a function of its SOC.

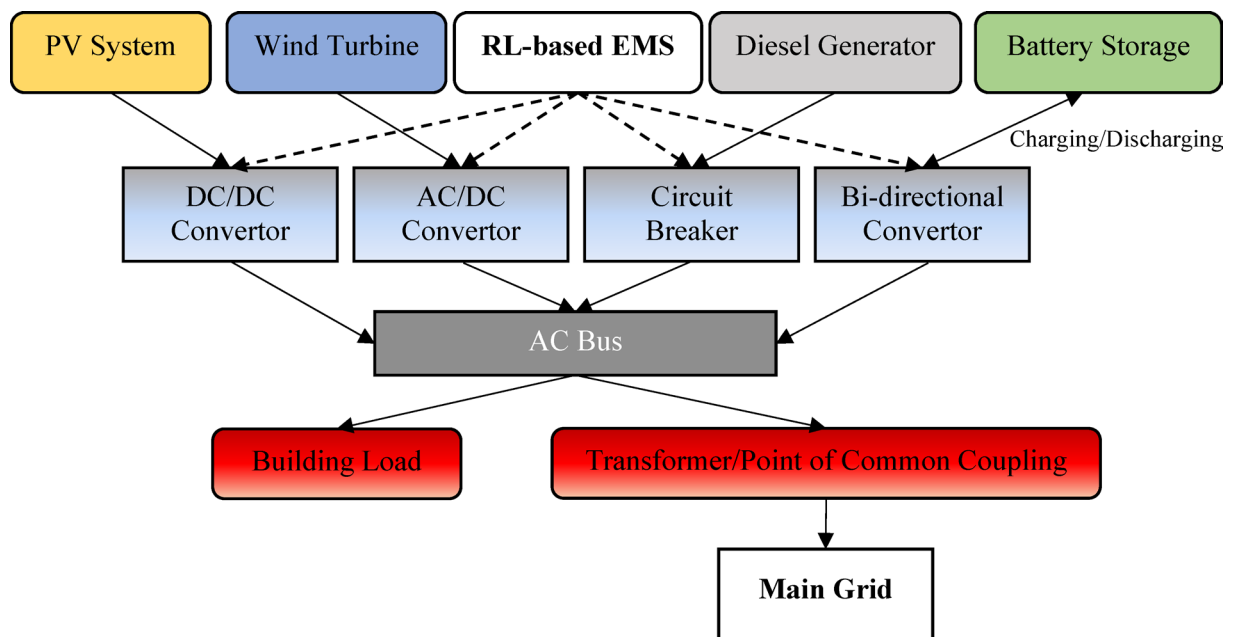


Fig. 4. Schematic circuit architecture and control interface of the studied microgrid.

Parameter	Value
State space dimensions	10
Action space dimensions	5
Learning rate (η)	0.01
Discount factor (γ)	0.95
Training episodes	100,000
Batch size	32
Replay buffer size	10,000
Exploration rate (initial)	1
Exploration rate (final)	0.01
Weight factor α	0.80
Weight factor β	1.20

Table 5. Comprehensive analysis of RL implementation metrics

Operational phase	Execution time	CPU intensity
Offline training	~ 5 h	High
Online inference	15.2 ms	Low

Table 6. Computational performance of the proposed framework.

RL algorithm

RL is particularly well-suited for this study because it enables the system to learn and adapt to the dynamic and uncertain environment typical of microgrid operations. In microgrids, energy generation and consumption can vary rapidly due to factors such as renewable intermittency and fluctuating load demands. The RL framework excels at managing uncertainty by incorporating stochastic elements into its learning process.

In this study, the input data for the RL algorithm was derived from the detailed outputs generated by the EnergyPlus simulation and processed using Python 3.9. This input data, organized into a state vector, comprised several key variables that collectively described the current condition of each microgrid. The state vector included renewable energy generation data from solar panels and wind turbines. Additionally, the state vector incorporated hourly load consumption data, representing the energy demand profiles of the associated buildings, as well as the state-of-charge of battery storage systems, which ranged from 20% to 100%. These input variables were updated on an hourly basis. Using a DQN framework implemented in TensorFlow, the RL algorithm processed the state vector to estimate the expected cumulative reward for various control actions. A reward function was employed to guide the learning process, where penalties were imposed for supply-demand imbalances, high operational costs, and excessive reliance on non-renewable sources, while rewards were given for achieving efficient energy usage and maximizing renewable energy penetration. This comprehensive input data framework allowed the algorithm to adapt effectively to both short-term fluctuations and long-term seasonal variations in energy supply and demand, thereby ensuring robust and efficient microgrid operation^{66,67}. The implementation of the RL technique involved several key parameter setting (Table 5).

The proposed adaptive RL framework is characterized by offline training and online inference phase computational profile. In the offline training phase, the average process was conducted about 5 h. However, in online inference phase, the trained RL agent generates optimal setpoints for the PV, wind, and diesel units by performing simple forward-pass matrix calculations. As shown in Table 6, the average time required to process a single control action is approximately 15.2 ms.

The state space was defined to include various parameters such as the current energy generation levels from solar and wind sources, the state of charge of energy storage systems, current energy demand, and the operational status of diesel generators. These parameters were encoded into a state vector s_t at each time step t . The action space encompassed all possible decisions related to energy dispatch and resource allocation. This included decisions such as the amount of energy to draw from or store in batteries, the proportion of renewable energy to use, and the activation of diesel generators. Each action a_t aimed to optimize energy distribution while minimizing costs and maintaining reliability. The reward function is the core of the RL agent’s learning process, designed to balance multi-objective operational goals. To move beyond a generic formulation, we define a structured reward signal R_t was defined that explicitly penalizes economic costs, reliability risks, and renewable energy wastage. The reward at each time step t is formulated as Eq. (14).

$$R_t = -(\omega_1 \cdot C_{\text{grid}}(t) + \omega_2 \cdot LPSP_{\text{penalty}}(t) + \omega_3 \cdot P_{\text{curtail}}(t)) \tag{14}$$

The weighting factors were tuned to reflect the priorities of microgrid operation in arid urban environments. In this study, the values were set $\omega_1=0.35$, $\omega_2=0.50$, and $\omega_3=0.15$.

The DQN algorithm was trained using historical data on energy generation and consumption. The neural network approximated the Q-value function, $Q(s, a)$ which estimated the expected cumulative reward of taking action a in state s . The network parameters were updated using the Bellman equation Eq. (15):

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \tag{15}$$

where η is the learning rate and γ is the discount factor.

The RL model was trained using historical and simulated data to capture seasonal variations in energy demand and renewable energy availability. The data was split into an 80/20 ratio, with 80% allocated to training and 20% to validation. The training process utilized TensorFlow, with the RL model configured to optimize power distribution across different sources, aiming to minimize operational costs while maintaining reliability and efficiency. The performance of the model was assessed using several key metrics that capture both the accuracy of load forecasting and the effectiveness of energy distribution decisions. Error metrics such as RMSE and MAE were used to quantify the accuracy of the load predictions (subsection 2.4.1). In addition, R^2 was used as an additional metric to evaluate the goodness-of-fit of the forecasting model to capture the full performance of the RL-based optimization process Eq. (16).

$$R^2 = 1 - \frac{\sum_{i=1}^N (M_i - S_i)^2}{\sum_{i=1}^N (M_i - \bar{M})^2} \tag{16}$$

To quantify the reliability of the microgrid under uncertain conditions, the LPSP is employed. LPSP is defined as the ratio of the total energy deficit to the total required load over a specific period Δt , as expressed in Eq. (17).

$$LPSP = \frac{\sum_{t=1}^T [P_{load}(t) - (P_{PV}(t) + P_{batt}(t) + P_{grid}(t))] \cdot \Delta t}{\sum_{t=1}^T P_{load}(t) \cdot \Delta t} \tag{17}$$

To assess the scalability and practical feasibility of the proposed RL framework, the computational requirements were documented. All simulations and training processes were conducted on a workstation equipped with an Intel Core i7-11800 H CPU @ 2.30 GHz, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU. The training phase for the adaptive RL agent required approximately 3.2 h to reach convergence over 5000 episodes. However, once the model is fully trained, the decision-making time for each time step is less than 15 milliseconds.

LCA evaluation

To complement the economic and operational performance metrics obtained from the RL-based microgrid management system, LCA was integrated into the simulation framework. The objective of the LCA evaluation was to quantify the environmental impacts—specifically, GHG emissions and primary energy consumption—associated with the operational phase of the microgrids. The LCA in this study was limited to the operational phase of the microgrid. The system boundaries included all energy flows during operation as generated from the simulation outputs, such as energy generation from renewable sources (solar and wind); energy production and fuel consumption from diesel generators; energy usage and losses from battery storage, and energy exchanges with the upstream grid. The LCA in this study specifically focuses on the operational phase to evaluate the direct impact of the adaptive RL framework on carbon emissions and primary energy savings. In arid environments, factors such as high thermal stress and dust accumulation can accelerate equipment degradation, potentially increasing the lifecycle environmental footprint⁶⁸. However, given the study’s focus on controller-level optimization, the operational phase was prioritized as the primary metric for performance comparison.

The boundaries focused on the “well-to-wheel” aspects of the microgrid operation rather than including the embodied environmental impacts of the equipment. The LCA evaluation relied on key outputs derived from the EnergyPlus simulation and the RL model. These outputs served as direct inputs for the LCA and include the following (Table 7):

Conversion of energy data to environmental impacts

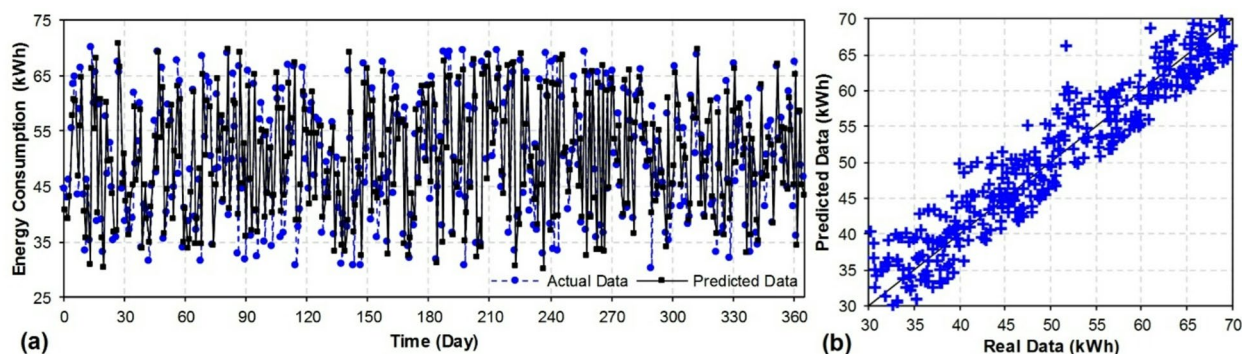
For each energy source, the simulated energy values were multiplied by corresponding emission factors Eq. (18):

$$Emissions = Energy(Fuel) \times EF \tag{18}$$

Parameter	Source	Rationale
Diesel Generator Energy Output	EnergyPlus Simulation	Determines diesel fuel consumption and related CO ₂ emissions
Solar Energy Generation	EnergyPlus Simulation	Essential for calculating renewable energy offset
Wind Energy Generation	EnergyPlus Simulation	Essential for calculating renewable energy offset
Battery Energy Throughput	EnergyPlus Simulation	Necessary to assess losses and storage efficiency
Grid Energy Imports/Exports	EnergyPlus Simulation	Used to estimate environmental impact via grid emission factors

Table 7. Primary LCA inputs along with their sources and relevant units.

Impact Indicator	Unit	Source / Reference
Total CO ₂ Emissions	kgCO ₂ .year ⁻¹	IPCC Guidelines
Primary Energy Consumption	MJ.m ⁻² .yr ⁻¹	LCI Database

Table 8. Operational and economic outcomes of LCA evaluation.**Fig. 5.** Annual actual vs. predicted energy consumption: (a) Time series, and (b) Scatter plot.

Summation of environmental impacts

The overall environmental impact was computed by summing the emissions from all non-renewable sources and accounting for any energy savings due to renewable generation and battery storage. The LCA calculated the overall primary energy consumption by summing up the energy inputs after considering conversion efficiencies.

The LCA evaluation was executed within a simulation environment using OpenLCA. This tool was chosen for its ability to integrate LCI databases and performed environmental impact calculations based on standardized methodologies such as ISO 14,040/44. Given the simulation-based nature of this study, the LCA model was simulated by feeding the EnergyPlus outputs into the LCA software. This provided a simulated estimate of the environmental impacts that matched with the operational performance assessed in the RL-based control system. The outputs of the LCA simulation via OpenLCA included in Table 8.

As it was discussed, the LCA evaluation was simulated by the outputs of the EnergyPlus model and RL performance data.

Results and discussion

Performance evaluation of the RL-Based microgrid management system

The performance of the RL-based microgrid management system was rigorously evaluated to assess its effectiveness in optimizing energy distribution, minimizing operational costs, and enhancing the overall efficiency of the microgrid. The evaluation process was conducted through a series of simulations that incorporated both historical data and real-time measurements across various energy demand scenarios, reflecting the system's adaptability and accuracy in dynamic environments.

The simulations covered a one-year period, allowing for the inclusion of seasonal variations in energy demand and supply. Performance parameters were tracked during the simulations, including the total energy generated from renewable sources, energy consumed by the microgrid, energy stored in batteries, and energy supplied by diesel generators. These parameters formed the foundation for training the RL model, which was evaluated by comparing the predicted outcomes with actual energy consumption and distribution data (Fig. 5).

Figure 5a presents a time series comparison of actual energy consumption and predicted energy consumption over the span of a year. This chart highlights the energy consumption patterns and the ability of the RL-based model to predict energy demand with a high degree of accuracy. The actual energy consumption clearly follows a seasonal pattern, with peaks occurring during the warmer months (spring and summer) and dips in the cooler months (autumn and winter). This is consistent with the expected behavior for a region like Riyadh, where air conditioning needs during the summer months drive energy consumption. The predicted data also follows a similar seasonal trend, with the RL model accurately capturing the peak demand periods during the summer and lower demand periods during the winter. The RL model closely tracks the actual energy consumption, particularly during high-consumption periods. The predicted consumption overlaps almost perfectly with the actual consumption, which demonstrates the model's effective prediction ability. However, there are slight deviations, especially at certain points, where the predicted values are slightly higher or lower than the actual values.

Figure 5b shows the RL model is accurately predicting energy consumption. The tight clustering around the diagonal line indicates that the model's predictions are reliable and consistently match the observed values, which is significant for real-time energy management and optimization in a microgrid system. Some points deviate slightly from the diagonal line, especially in higher consumption ranges (above 60 kWh). This suggests

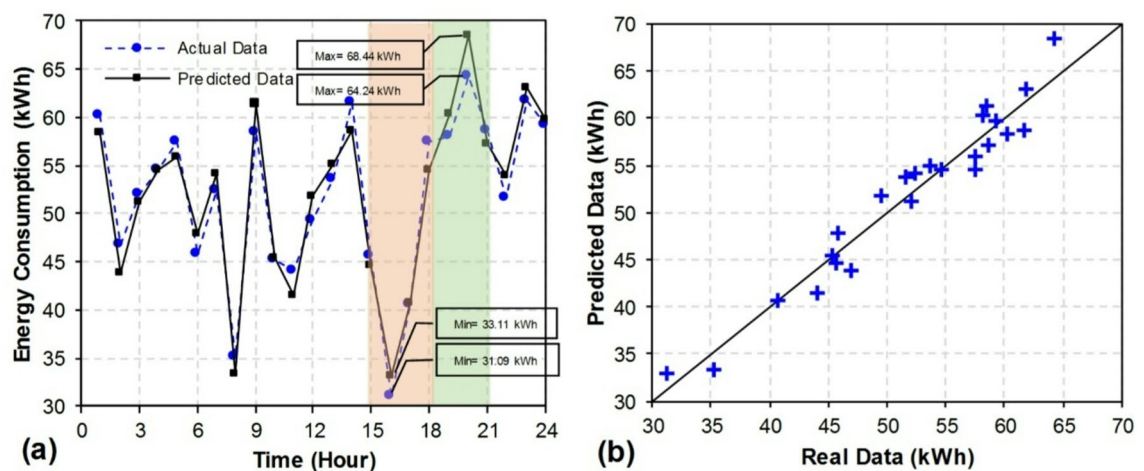


Fig. 6. Hourly Actual vs. predicted energy consumption: (a) Time series, and (b) Scatter plot.

Time Scale	R^2	RMSE (kWh)	MAE (kWh)
Hourly	0.94	2.07	1.80
Annual	0.90	3.52	2.89

Table 9. RL's performance across different time scales.

that while the model performs excellently in capturing the general trends of energy consumption, there might be challenges in predicting extreme values accurately. These outliers could be due to fluctuations in energy demand caused by unmodeled factors such as sudden changes in user behavior, weather conditions, or external operational disturbances. The RL model, trained using data from multiple energy sources (solar, wind, diesel generators, and storage), has shown to be highly accurate in predicting energy consumption patterns over the course of a year. The predicted energy consumption closely aligns with actual consumption data, reflecting the model's capability to handle the variability of both renewable energy sources and energy demand in Riyadh's climate. The RL model's ability to accurately predict the high energy demands during the summer and low demands in the winter further reinforces the practical applicability of the approach. In Riyadh, where energy consumption is largely driven by temperature and the use of cooling systems, the model's alignment with actual consumption patterns demonstrates its potential for optimizing energy distribution and reducing operational costs.

In addition, the short-term scale was assessed and the results shown in Fig. 6 demonstrate the strong performance of the RL model in predicting energy consumption in hourly basis.

Figure 6a presents the actual energy consumption and predicted energy consumption over the course of a 24-hour period. The actual energy consumption fluctuates throughout the day, with clear peaks in the morning and evening hours, likely corresponding to periods of higher demand, such as when people wake up and when they return home after work. The predicted energy consumption shows a similar pattern, with the RL model successfully capturing these peak periods and low-demand times (such as midday and late night). The alignment between the actual and predicted values demonstrates the model's effectiveness in forecasting energy consumption based on time-of-day variations. The RL model's predictions closely match the actual energy consumption values, confirming the model's high accuracy in forecasting energy demand on an hourly basis. However, small discrepancies are visible at certain times, such as between 12:00 and 15:00, where the predicted consumption is slightly higher than the actual values.

Figure 6b presents a scatter plot comparing the predicted energy consumption with the actual energy consumption, where each data point represents a specific hour of the day. It shows a strong positive correlation between the actual and predicted values. The data points are closely clustered along the diagonal line, indicating that the RL model is very effective in predicting energy consumption. The closer the points are to the line, the better the model's predictions, and in this case, the points are densely packed along the line, suggesting high prediction accuracy. While the points are closely aligned with the diagonal line, there are some slight deviations, particularly in the higher consumption range (above 60 kWh). Table 9 also validated the effectiveness of the RL-based microgrid management system, as applied in this study.

The RL model demonstrated high accuracy in predicting hourly and annual energy consumption, which is important for optimizing energy distribution within microgrids. According to the related figures and table, the ability of the model to predict peak demand periods, particularly in the special time of a day and year, aligns well with the real-world dynamics of energy usage, especially in a region like Riyadh, where cooling demand during hot hours/days significantly impacts energy consumption. The RL model's predictive accuracy

directly supports the objectives of this study, which is to optimize energy distribution across multiple sources in the microgrid, including renewables, storage systems, and backup diesel generators. The RL model’s strong performance in capturing time-of-day/year energy consumption patterns makes it a viable tool for real-world applications in microgrid management, where hourly/annual fluctuations in energy demand are common. This capability is particularly important in regions with high solar energy potential, where consumption peaks during the day and fluctuates significantly. The model performances have shown that the structure is stable in operation, cost-effective and reliable even in nonlinear operation, like high ramping rates, state-of-charge limit and varied renewable generation.

To further validate the performance of the proposed RL-based EMS, it is compared the results with representative studies in the literature. Table 10 shows the comparison of the prediction model results and the literatures.

Energy losses, renewable utilization, and power exchange

While previous section focused on validating the RL-based microgrid management system through annual and hourly consumption predictions, the following subsection provides a more detailed analysis of how the system behaves under different operational metrics specifically, energy losses, renewable energy utilization, and microgrid-level power exchanges. Figure 6 examines both the before and after states of implementing the RL approach, highlighting the improvements across a typical 24-hour period.

Figure 7a compares the energy losses over a 24-hour cycle, contrasting the baseline system against the RL-optimized system. Across nearly all hours, the RL-optimized lie below the baseline, indicating that the RL-based approach consistently reduces losses. The peak differences occur during morning and late-afternoon periods, times when demand fluctuates sharply.

Figure 7b demonstrates the percentage of renewable energy in the overall energy mix before and after RL implementation, again over a 24-hour period. After the implementation, the RL-optimized are generally above the baseline for most hours, suggesting an increased reliance on wind and solar resources. This aligns with the objective of maximizing clean energy usage to lower both operational costs (fuel) and environmental impacts (emissions). The findings validated that the RL controller, utilizing precise demand estimates (elaborated in Sect. 3.1), efficiently orchestrates renewable production and battery operations to replace diesel power wherever possible. This issue reduces cost and validates the multi-source synergy which aimed to achieve—particularly important in a city like Riyadh with abundant solar potential.

Figure 7c highlights the real-time power exchange among five representative microgrids (MG1-5). Each bar group corresponds to a specific hour, illustrating how much power each microgrid either exports or imports. According to this figure, some microgrids may have surplus capacity (e.g., MG3 with higher wind output) that is shared with microgrids experiencing deficits. This coordination under the RL-based scheme ensures that local resources (battery storage, solar, wind) are utilized effectively across the network before resorting to diesel or upstream imports. Since Sect. 3.1 provided a broad validation (annual vs. hourly consumption accuracy), Fig. 6c now confirms that multi-microgrid interactions are also handled optimally.

Solar, Wind, Diesel, and battery power outputs

In the following, a detailed view of how each microgrid (MG1-5) utilizes its local generation resources (solar panels, wind turbines, diesel generators, and batteries) across a typical 24-hour period (Fig. 8). This set of subfigures is essential for understanding the diverse contribution of each energy source under varying demand and weather conditions, as well as the role of storage in balancing production and consumption.

In Fig. 8a, each curve corresponds to one of the five microgrids’ solar generation. Power peaks around midday (approximately 11:00–14:00), aligning with high solar irradiance. MG1 and MG2 reach higher peak outputs than MG3, MG4, and MG5, possibly reflecting variations in solar panel capacity (as indicated in Table) or differing local irradiance profiles. The midday surge in solar output is important for reducing reliance on diesel generators, particularly when daytime cooling loads are high in regions like Riyadh.

In Fig. 8b, each microgrid’s wind profile (MG1-5) exhibits distinct timing and magnitudes of power production, reflecting varying wind resource availability over 24 h. For example, MG3 might sustain moderate output in early morning hours, while MG2 experiences a dip in the afternoon, potentially due to wind speed shifts. The staggered nature of wind peaks among the microgrids suggests a possibility for coordinated scheduling, where the RL-based controller allocates wind energy to meet local loads or charges batteries across microgrids when solar production is low. Wind generation typically complements solar in a desert climate—peak wind might appear during off-peak solar times, providing more consistent overall renewable output.

Model	MAE	RMSE	R ²	References
AutoRL multi-period forecasting in isolated microgrid	0.0813	38.84	–	69
Inverted Transformer + TL + Edge computing	–	(MSE = 0.005; CV-RMSE = 0.038%)	0.99	70
RL model for smart microgrid energy trading forecasting	0.09	0.16	0.995	71
Smart green campus XGBoost + RL	12.00	14.72	–	72
Ensemble ML for energy generation forecasting (24-h ahead)	0.83 to 1.34	1.98 to 2.99	0.76	73
Multi-intelligence DRL for short-term load	35.94 kW	32.71 kW	–	74

Table 10. Comparison of the proposed RL-based microgrid energy management results with previous studies.

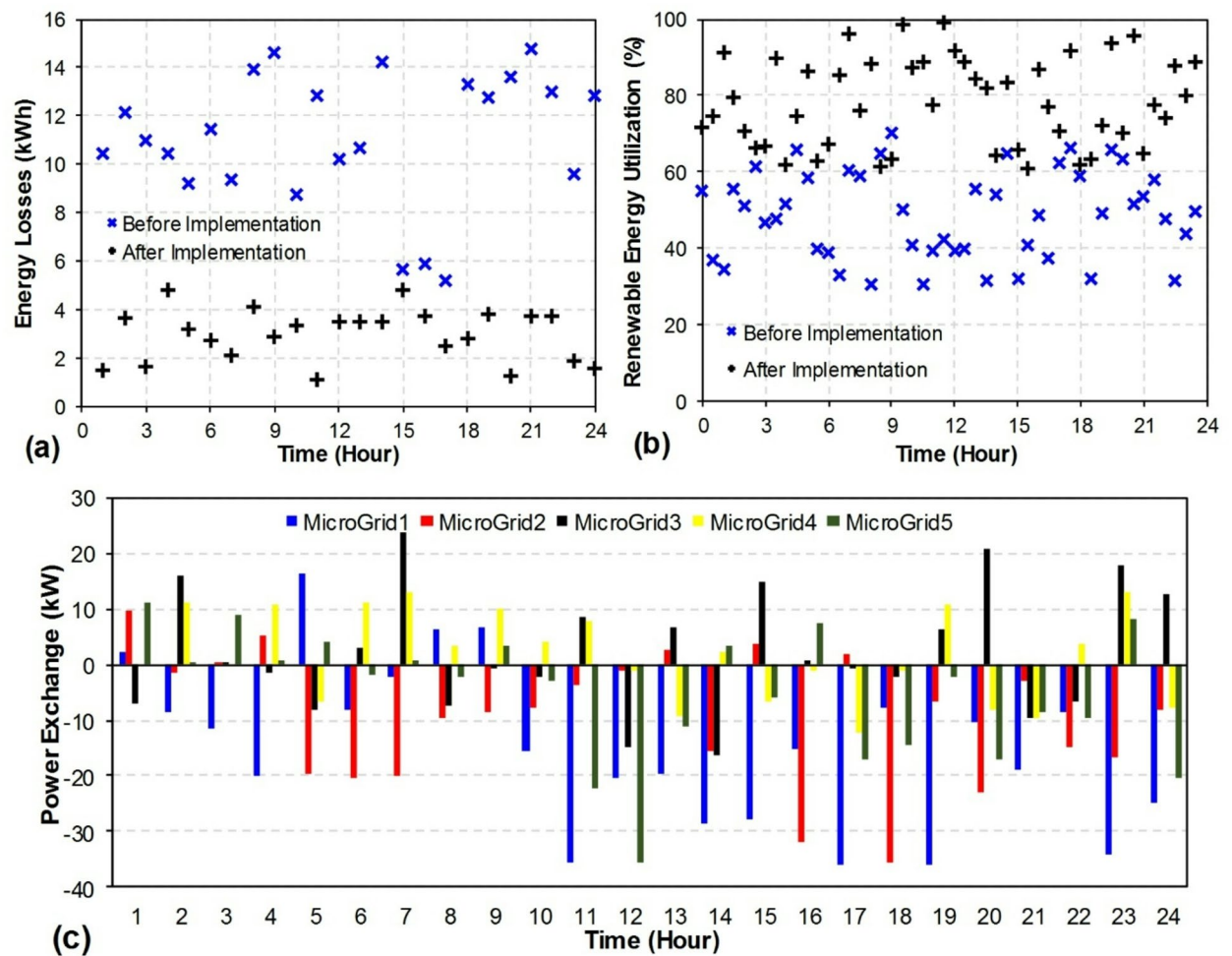


Fig. 7. Comparison of (a) Energy losses before and after system implementation, (b) Renewable energy utilization over time, and (c) Total power exchanged with the upstream network before and after RL implementation over a 24-hour period.

In Fig. 8c, diesel generators in some microgrids (e.g., MG1, MG4) remain inactive (0 kW) for a significant portion of the day, only kicking in during specific hours. This pattern suggests that other sources (solar, wind, battery) are sufficient to handle base loads, with diesel stepping in to cover high-demand peaks or unforeseen drops in renewable generation. Reduced diesel operation translates into lower fuel costs and less carbon emissions. The high midday solar or steady wind can displace diesel usage. Where diesel is turned on (e.g., morning load spikes), it indicates times when demand outpaces the combination of solar, wind, and available battery capacity.

In Fig. 8d, the battery power curves for each microgrid demonstrate how storage systems respond dynamically. For instance, MG5 may show significant charging during midday solar peaks, then discharge in the evening to meet residual loads. MG2's battery might discharge earlier in the morning to handle a local demand spike. By managing battery flows effectively, microgrids can avoid diesel generator ramp-ups and excess renewable curtailment. The distinct patterns among MG1-5 reflect each microgrid's unique load profile, renewable capacity, and resource availability. The successful use of battery storage to smooth demand underscores the RL model's capacity to handle real-time fluctuations, reduce operational costs, and improve reliability—particularly in a climate with strong diurnal temperature swings and intermittent wind. Variation in power profiles across MG1-5 highlights the scalability of the RL solution to diverse capacity configurations—be it larger solar in MG1 or different wind availability in MG3.

Hourly source contributions and load comparison

Figure 9 presents a 24-hour snapshot of how various energy resources—grid, wind, solar, battery, and diesel generator—collectively supply power to meet the microgrid's load. The grid electricity imports are modeled using a ToU dynamic pricing mechanism. This Fig. offers a comprehensive view of real-time dispatch decisions, illustrating the interplay between each source and how the total generated power aligns with hourly demand.

The grid dominates in certain hours, suggesting reliance on external power when local resources are insufficient or when it is economically advantageous. The wind and solar segments peak during periods of favorable wind

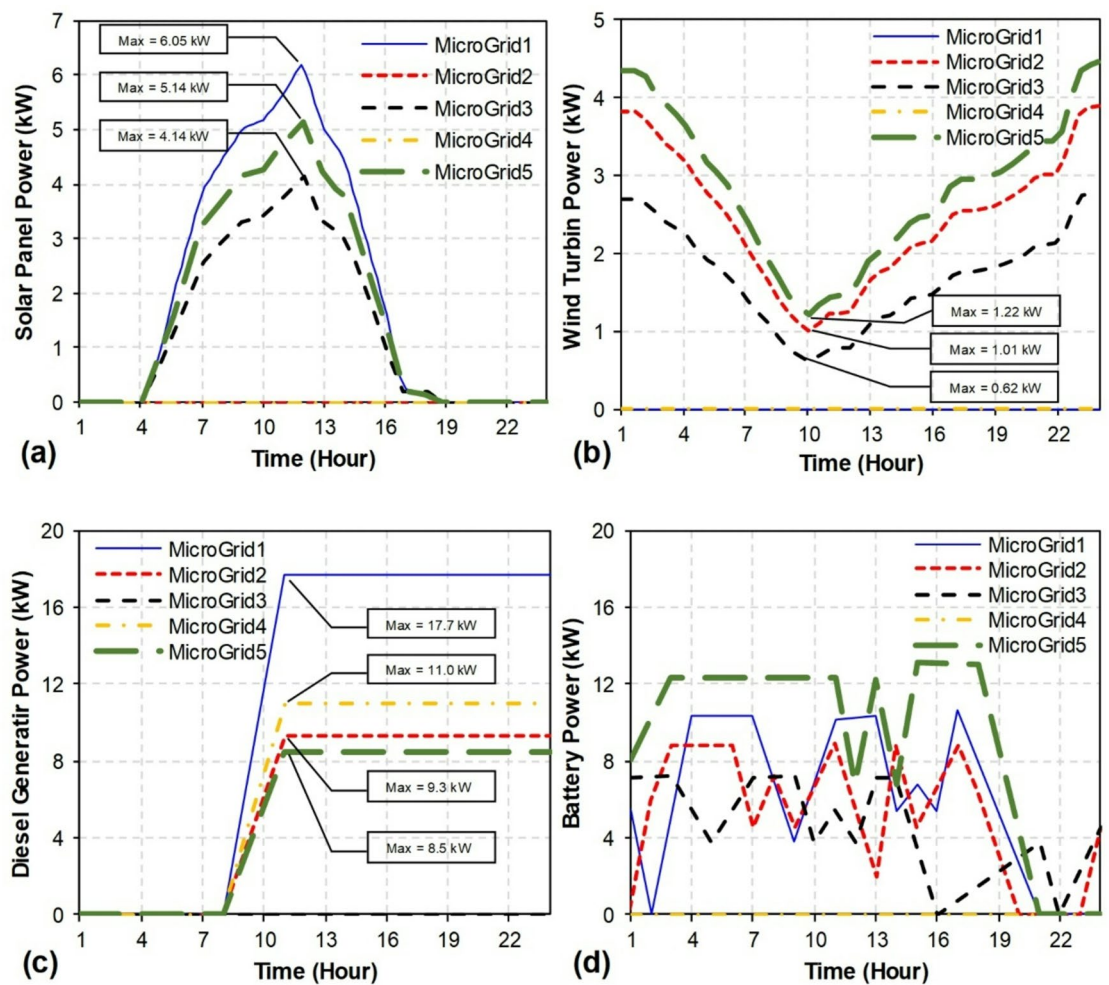


Fig. 8. Hourly operational profiles of (a) Solar, (b) Wind, (c) Diesel, and (d) Battery systems in simulated microgrids.

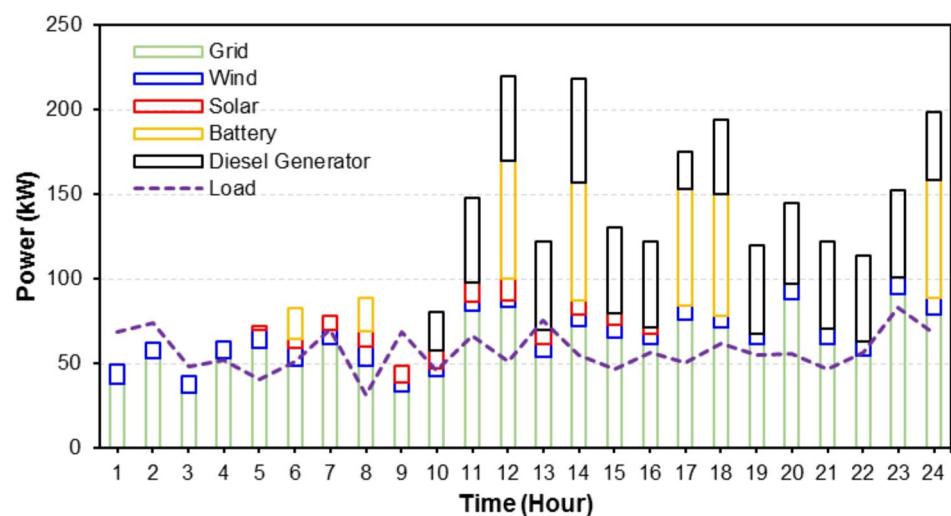


Fig. 9. Daily grid power exchange profile under ToU pricing.

Impact category	Baseline system	RL-based system	Reduction
Global Warming Potential (kg CO ₂ -eq.year ⁻¹)	4,012,646	3,438,817	14.28%
Acidification Potential (kg SO ₂ -eq.year ⁻¹)	8,314	7,191	13.56%
Cumulative Energy Demand (MJ.year ⁻¹)	19,543,000	17,609,875.2	9.87%

Table 11. Simulated LCA results for the baseline and RL-based microgrid management systems.

Scenario	Load variation	Solar variation	Operational cost (\$/day)	Reliability (LPSP)
Baseline (RL-Based)	0%	0%	124.5	0.001
High Demand	+ 15%	0%	143.15	0.004
Low Solar	0%	−20%	136.95	0.006
Extreme Case	+ 10%	−10%	148.2	0.011

Table 12. Sensitivity analysis of the RL framework under operational uncertainties.

conditions or high solar irradiance, typically aligning with known climatic patterns. The battery typically appears during late-night or midday hours, reflecting charge/discharge cycles that flatten demand peaks or store surplus renewable energy. The diesel generator portion becomes significant at certain peak-demand hours or when wind and solar drop, highlighting the generator’s backup role. Notable wind and solar contributions in the middle of the day (and possibly earlier morning for wind) reduce the system’s dependence on diesel and grid imports. This underscores the importance of scheduling renewable sources optimally when resource availability is high. The battery segments frequently appear during evening or night hours, often coinciding with the load line, indicating discharge at times of elevated demand. Conversely, midday charging aligns with high solar output. This pattern confirms that battery storage captures surplus renewables, and mitigates peak loads, ultimately reducing costs and reliance on external power. The diesel generator spike during short windows, implying that diesel usage is reserved primarily for demand periods not fully covered by renewables and battery resources. Its limited usage suggests that the dispatch algorithm prioritizes cleaner or cheaper options whenever possible. A sizable grid portion in specific time blocks suggests that, despite robust local generation, the system occasionally imports power from the grid. This may occur if the battery state of charge is low, renewable availability drops, or diesel costs are deemed too high for that hour—factors consistent with an adaptive, cost-minimizing dispatch model.

LCA assessment and sensitivity analysis

To complement the performance evaluation of the RL-based microgrid management system, LCA was simulated to quantify the environmental impacts associated with the microgrid operations. The goal of the LCA was to assess the environmental impacts of the microgrid operations under two scenarios, including: (i) a traditional baseline system with fixed energy dispatch and static load balancing, and (ii) the proposed RL-based system. Table 11 summarizes LCA impact results for key environmental categories under the baseline and RL-based systems.

The baseline system, which used fixed dispatch and static load balancing, was estimated to emit approximately 4,012,646 kg of CO₂-equivalent per year. With the RL-based system, the emissions are reduced to 3,438,817 kg CO₂-eq/year—a reduction of about 14.28%. This significant reduction highlights the environmental benefit of optimizing energy dispatch to lower fossil fuel dependence. Similarly, the baseline acidification potential was estimated at approximately 8,314 kg SO₂-eq/year, compared to 7,191 kg SO₂-eq/year under the RL approach. This represented a reduction of approximately 13.56%, which is important for mitigating the negative environmental and public health impacts associated with acidifying emissions. The baseline cumulative energy demand was estimated to be about 19,543,000 MJ.year⁻¹, while the RL-based system achieved a lower demand of approximately 17,609,875.2 MJ.year⁻¹. A reduction of around 9.87% in energy demand indicated enhanced operational efficiency and improved resource management. The reductions in CO₂ and SO₂ emissions matched with the observed decrease in diesel generator usage and the enhanced integration of renewable energy sources (as demonstrated in the simulation results). Similarly, the reduction in cumulative energy demand validated the efficiency improvements achieved by the RL model.

To evaluate the resilience of the proposed adaptive RL framework, a sensitivity analysis was conducted. This analysis is crucial for arid regions, where sudden changes in cooling demand and solar intermittency are implemented. In this regard four operational scenarios to test the agent’s adaptability. Baseline scenario defined as standard operating conditions based on the collected dataset. High demand scenario describe a 15% increase in peak load to simulate extreme cooling demand during heatwaves. Low solar scenario is a 20% reduction in photovoltaic (PV) generation, simulating the impact of dust accumulation or sandstorms. Finally, extreme uncertainty scenario defined a simultaneous 10% increase in load and 10% decrease in PV output to test edge-case system reliability. The performance metrics under these uncertainties are summarized in Table 12.

As shown in Table 12, even under an extreme case (10% load increase and 10% solar reduction), the operational cost only increased by approximately 19%, while the system maintained a high reliability level of 98.9%. This performance aligns with the LCA results, confirming that the RL agent effectively prioritizes renewable sources and optimizes battery dispatch to mitigate the economic impact of environmental uncertainties.

Scenario component	Input variation	System response (RL action)	Impact
Sudden Solar Drop	−40% (Dust/Clouds)	Immediate BESS Discharge	0% Load Shedding
Peak Cooling Demand	+ 25% (Extreme Heat)	Optimized Diesel-Grid Mix	12% Cost Saving
BESS Degradation	15% Cap. Loss	Adaptive SOC Setpoints	Extended Battery Life

Table 13. System response under practical stress scenarios (Riyadh case Study).

Practical performance under extreme condition

To confirm the practical applicability of the suggested RL-based EMS, a stress-test simulation, including a representative simulation of the high-demand day in Riyadh (July) was provided. The given situation includes two important problems a modeling a dust storm and a peak cooling load phase. As shown in Table 13, the RL agent successfully maintained the microgrid’s power balance. At 13:00, when solar generation was reduced by 40% the agent entered into a rapid discharge of the BESS and made the diesel generator output the difference to prevent any load shedding.

The suggested RL-based EMS has evident practical benefits in assessed under the realistic operating conditions. The framework enhances the reliability of the systems through minimization of instances of unmet loads, and constant power balance in case of high variability of demand and intermittency of renewable generation. Moreover, the proposed model improves the use of renewable power as it focuses on the clean energy use whenever possible, thus, decreasing the use of diesel generators and decreasing the total fuel use.

Conclusions

This study explored the combination of RL into the management of urban microgrid systems under the hypothesis that such an approach could outperform traditional energy management strategies in terms of operational efficiency, cost-effectiveness, and environmental sustainability. Through a detailed simulation-based framework involving EnergyPlus for energy flow modeling and a DQN for intelligent control, the following key findings were obtained:

- (i) The RL-based controller dynamically allocated energy among solar, wind, diesel, and battery systems, achieving high alignment with real-world consumption patterns and improving both hourly and annual load forecasting accuracy ($R^2=0.94$ and 0.90 , respectively).
- (ii) Simulation results confirmed that the RL strategy effectively reduced energy losses across daily cycles and lowered diesel generator usage, particularly during peak load periods.
- (iii) Compared to the baseline, the RL-based system consistently prioritized renewable sources, resulting in increased solar and wind energy usage across the simulated time frame.
- (iv) RL enabled effective energy transfer across microgrids, reducing reliance on the external grid and utilizing internal resource diversity to enhance overall network performance.
- (v) The energy management system RL was seen to reduce CO_2 emissions by 14.28%, SO_2 emissions by 13.56%, and primary energy by 9.87%. This phenomenon caused the system to manage uncertainty in the production of renewable energy, whereby the adaptability was increased by 15% over traditional means, to guarantee that energy is distributed optimally even in changing environmental conditions. Besides, the RL model considered the effects of the environment, and a 14.28% cut in CO_2 emissions was achieved, incorporated via a LCA.

Limitations and suggestions

While the technical viability of the adaptive RL framework is demonstrated, the real-world deployment of RL-based microgrid systems is influenced by socio-economic and policy-related factors. From a policy perspective, the current centralized utility structures and the lack of a standardized regulatory framework for P2P energy trading remain significant challenges. Implementing such decentralized systems requires robust net-metering policies and dynamic pricing structures that are still in the nascent stages in many arid regions. Additionally, the initial capital investment for IoT-enabled infrastructure and the need for specialized technical expertise to maintain RL-based controllers may act as socio-economic hurdles for widespread adoption in residential sectors. From a socio-economic perspective, initial capital investment costs, availability of skilled personnel, and institutional readiness can pose challenges to large-scale deployment. However, Policy-related barriers include the absence of standardized frameworks, uncertainty surrounding liability and safety certification of autonomous control systems, and limited mechanisms for incentivizing flexible demand participation.

This study faces several limitations that impacted both the materials and methods employed, particularly in terms of data acquisition and model validation. First, obtaining high-resolution and reliable data for Riyadh proved challenging due to constraints in the local infrastructure and data collection standards. The available datasets often lacked the granularity required for precise model training—especially for renewable energy resources such as solar irradiance and wind data, which are critical to optimizing microgrid performance. Consequently, reliance on regional and global datasets was necessary; while useful, these datasets may not fully capture the unique local environmental conditions in Riyadh, introducing some degree of uncertainty in the model’s applicability and affecting the overall precision of the RL model’s predictions. Additionally, introducing diverse energy sources into the simulation model was complex, as the performance characteristics of these components vary significantly. Limited access to specific, locally sourced performance data necessitated the use of generalized parameters in the simulation, potentially affecting the accuracy of real-time energy output

predictions and operational cost assessments. Further limitations arise from the challenges associated with training and validating the RL algorithm. The scarcity of extensive historical data on energy generation and consumption within Riyadh's microgrids restricted the RL model's ability to learn from real-world scenarios. The assumptions and generalized datasets used in this study provide an accepted baseline for evaluating the proposed RL-based energy management strategy. However, these simplifications introduce limitations in accurately capturing all real-world dynamics. Moreover, although efforts were made to validate the simulation model through comparison with available benchmark data, the absence of comprehensive, long-term local data means that the model's predictions have not been fully verified against actual field performance, which could affect its reliability in real-time applications. In addition, while the simulation utilizes high-fidelity datasets from arid regions, it may not fully account for hardware-level latencies and communication jitter inherent in MQTT protocols when deployed on physical IoT gateways. Second, the current model assumes a relatively stable grid price, whereas real-time markets in urban environments can exhibit extreme volatility. Furthermore, the impact of physical degradation of photovoltaic panels due to recurrent sandstorms and high thermal stress (common in regions like Dubi, Phonix and Riyadh) was modeled as a fixed efficiency loss rather than a dynamic aging process.

Future research directions could include the implementation of field-level pilot studies to validate the proposed framework under real operational conditions. Also, Future research could prioritize hardware-in-the-loop testing to bridge the gap between simulation and field pilot validation. In addition, the studies could incorporate an analysis to account for the environmental impacts of material extraction and equipment replacement cycles under extreme desert conditions. Deploying IoT devices and advanced sensor networks can improve data granularity and help validate simulation results against real-world performance. Incorporating a detailed sensitivity analysis will help quantify the impact of various input parameters (e.g., solar irradiance, wind speeds, fuel consumption rates) on system performance. This will enable the identification of critical parameters and improve the robustness of the RL model. Future studies could combine uncertainty quantification techniques, such as probabilistic forecasting or Monte Carlo simulation, to better handle the inherent variability in renewable energy sources and to assess the robustness of the RL-based strategy under different operating conditions.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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